16S LiFePO4 BMS (48V / 51.2V) : SP Road Procurement Checklist

FIELD PROCUREMENT SHEET
BANGALORE OTC GUIDE
Total Items: 124
Verified Domestic Scope

Organization: Subsystem-by-subsystem hierarchy with dedicated sections for Non-ET Store vs ET Store items.
Features: Blank 'SP Road Store' column for handwritten shop entries, design vs purchase quantities, specs/alternatives, and quote fields.

PART 1 : COMPONENTS SOURCED FROM OTHER DISTRIBUTORS (NON-ET STORE) - ORGANIZED BY SUBSYSTEM

[✓]	Subsystem	Ref Des	Component / Value	Package / Footprint	Req / Buy	Technical Specifications & Counter Substitutions	SP Road Store (Write Shop Name)	Quote (₹) (Write Price)
	SUBSYSTEM : EXTERNAL & TOP LEVEL INTERCONNECT (8 items)							
[ ]	External	RS1	FL-2B 100A 75mV	Chassis_BoltMount	1 req 1 buy	Specs: XIQI Electric CG FL-2B 100A 75mV DC Current Shunt Resistor Alts/Notes: FL-2C 100A 75mV with base (Robu ₹499), FL-2 150A 75mV	[ ]	Est: ₹352.0 [ ]
[X]	External	R_PRE	50 Ω 25W	Chassis_Mount_55mm_P..	1 req 1 buy	Specs: 50 Ohm 25W Aluminum Housed Wirewound Power Resistor (55mm mounting pitch) Alts/Notes: 50 Ohm 50W Aluminum Resistor (Sharvi ₹184.88) ALREADY IN HAND: Procured 50 Ω 25W aluminum chassis resistor (55mm pitch). Do not buy.	IN HAND	Procured
[ ]	External	P20	DB25 Male Solder Plug + Hood	Cable_Mount	1 req 1 buy	Specs: DB25 Male Solder-Type Connector Plug with Plastic/Metal Protective Shell Alts/Notes: DB25 Male Solder Cup (₹18.90) + DB25 Plastic Cover (₹18.00)	[ ]	Est: ₹35.0 [ ]
[ ]	External	P30	DB50 Male Solder Plug + Hood	Cable_Mount	1 req 1 buy	Specs: DB50 50-Pin D-Sub Male Solder-Type Connector Plug with Protective Dust Cover Alts/Notes: Rajiv DB50 Male Solder Type (₹82.60) + DB50 Metal/Plastic Hood	[ ]	Est: ₹95.0 [ ]
[ ]	External	J30	DD50_Female	THT	1 req 1 buy	Specs: 50-Pin D-Sub High-Density Female PCB Connector (Cell Taps) Alts/Notes: Rajiv DB50 Straight Female, NorComp 180-050-203L001	[ ]	Est: ₹112.1 [ ]
[ ]	External	J32	FROM SHUNT	MountingHole_8.4mm_M..	1 req 5 buy	Specs: FROM SHUNT High-Current M5/M6 Direct Bolt-Down PCB Pad Terminal Alts/Notes: M5/M6 SS304 Hex Bolt, Spring Washer, Flange Nut	[ ]	Est: ₹12.0 [ ]
[ ]	External	J33	PACK MINUS	MountingHole_8.4mm_M..	1 req 5 buy	Specs: PACK MINUS High-Current M5/M6 Direct Bolt-Down PCB Pad Terminal Alts/Notes: M5/M6 SS304 Hex Bolt, Spring Washer, Flange Nut	[ ]	Est: ₹12.0 [ ]
[ ]	External	J34	PACK PLUS TAP	MountingHole_8.4mm_M..	1 req 5 buy	Specs: PACK PLUS TAP High-Current M5/M6 Direct Bolt-Down PCB Pad Terminal Alts/Notes: M5/M6 SS304 Hex Bolt, Spring Washer, Flange Nut	[ ]	Est: ₹12.0 [ ]
	SUBSYSTEM : 01 CELL AFE (2 items)							
[ ]	01 Cell AFE	U1	BQ76952PFBR	QFP50P900X900X120-48N	1 req 1 buy	Specs: 16S LiFePO4 Battery Monitor, Protector & Balancer IC Alts/Notes: TI BQ76952PFBR, BQ7695202 (I2C default) AFE IC: Rare OTC. If not at Vishal/Syscom, use confirmed LionCircuits order.	[ ]	Est: ₹665.3 [ ]
[ ]	01 Cell AFE	Q21	BCX56-135	SOT-89-3_Handsoldering	1 req 1 buy	Specs: 80V 1A Medium Power NPN Transistor, SOT-89-3 Alts/Notes: BCX56-16, BCP56, 2SD882	[ ]	Est: ₹11.4 [ ]
	SUBSYSTEM : 02 POWER SUPPLY (2 items)							
[ ]	02 Power Supply	U10	LM5164DDA	HSOP-8-1EP_3.9x4.9mm..	1 req 1 buy	Specs: 100V 1A Synchronous Step-Down Buck DC-DC Converter Alts/Notes: 7Semi LM5164 Breakout, TI LM5017MR, LM5007 (100V step-down)	[ ]	Est: ₹548.7 [ ]
[ ]	02 Power Supply	D3	SMBJ60CA	D_SMB	1 req 1 buy	Specs: 60V 600W Bidirectional TVS Diode, SMB Alts/Notes: SMBJ64CA-13-F (₹31, In Stock Robu), SMCJ60CA, 1.5SMC62CA	[ ]	Est: ₹15.0 [ ]
	SUBSYSTEM : 03 MCU CONTROL (1 items)							
[ ]	03 MCU Control	U21	FM24CL64B-G	SOIC-8_3.9x4.9mm_P1...	1 req 1 buy	Specs: 64 Kbit (8Kx8) High-Endurance Non-Volatile I2C FRAM Alts/Notes: FM24CL16B (16Kb), FM24W256 (256Kb), Microchip 24LC256 (EEPROM) FRAM: Rare OTC. If not at Vishal, use Robu order. (24C64 EEPROM only for testing).	[ ]	Est: ₹232.0 [ ]
	SUBSYSTEM : 04 CURRENT SWITCHING (1 items)							
[ ]	04 Current Switc..	U2, U3	UCC27517DBVR	SOT-23-5	2 req 2 buy	Specs: 4A High-Speed Low-Side Gate Driver, SOT-23-5 Alts/Notes: UCC27511DBVR, FAN3100, MCP1407	[ ]	Est: ₹35.0 [ ]
	SUBSYSTEM : 06 BALANCE STATUS (1 items)							
[ ]	06 Balance Status	J20	DB25_Female	25_Socket_Horizontal..	1 req 1 buy	Specs: 25-Pin D-Sub Female Right-Angle PCB Connector (NTC Taps) Alts/Notes: Rajiv DB25 Female, Amphenol D-sub 25	[ ]	Est: ₹26.0 [ ]
	SUBSYSTEM : 07 ISOLATED CAN & COMMS (4 items)							
[ ]	07 Isolated CAN ..	U13	ISO1050DUBR	SOP-8_6.62x9.15mm_P2..	1 req 1 buy	Specs: 5V Isolated CAN Transceiver, SOP-8 Alts/Notes: TI ISO1050DUB, Robu.in ISO1050 Module, MAX14882	[ ]	Est: ₹230.1 [ ]

[✓]	Subsystem	Ref Des	Component / Value	Package / Footprint	Req / Buy	Technical Specifications & Counter Substitutions	SP Road Store (Write Shop Name)	Quote (₹) (Write Price)
[ ]	07 Isolated CAN ..	U14	B1209S-1WR3	Murata_MEE1SxxxxSC_THT	1 req 1 buy	Specs: 1W 12V-to-9V Isolated DC-DC Converter for Gate Drive Rail Alts/Notes: K-CUT B1209S-1W, Hi-Link HLK-B1209S-2WR3 (₹199, Robu)	[ ]	Est: ₹109.0 [ ]
[ ]	07 Isolated CAN ..	D5	PESD2CAN	SOT-23	1 req 1 buy	Specs: CAN Bus Dual-line ESD Protection TVS Diode, SOT-23 Alts/Notes: PESD1CAN, NUP2105L, ESDCAN01-2BWY	[ ]	Est: ₹12.0 [ ]
[ ]	07 Isolated CAN ..	D20	SMAJ6.0A	D_SMA	1 req 1 buy	Specs: 6.0V 400W Fast Unidirectional TVS Diode, SMA Alts/Notes: SMAJ5.0A, SMAJ6.5A, SMBJ6.0A	[ ]	Est: ₹4.2 [ ]
	SUBSYSTEM : 08 PRECHARGE CIRCUIT (2 items)							
[ ]	08 Precharge Cir..	D40	SMAJ75A	D_SMA	1 req 1 buy	Specs: 75V 400W Fast Unidirectional TVS Diode (Precharge Inductive Clamp, SMA) Alts/Notes: SMAJ70A, SMAJ78A, SMCJ75A (1.5kW), Robu.in 75V TVS	[ ]	Est: ₹3.3 [ ]
[ ]	08 Precharge Cir..	D42	BAT54S	SOT-23	1 req 1 buy	Specs: Dual Schottky Barrier Diode Series Pair (Precharge Sense ADC Rail Clamp, SOT-23) Alts/Notes: BAT54A, BAT54C, MMBD4148 dual, 2x 1N5819	[ ]	Est: ₹2.0 [ ]

PART 2 : COMPONENTS SOURCED FROM ET STORE (etstore.com) - ORGANIZED BY SUBSYSTEM

[✓]	Subsystem	Ref Des	Component / Value	Package / Footprint	Req / Buy	Technical Specifications & Counter Substitutions	SP Road Store (Write Shop Name)	Quote (₹) (Write Price)
	SUBSYSTEM : EXTERNAL & TOP LEVEL INTERCONNECT (2 items)							
[ ]	External	TH1..TH16	10k NTC 3950 (with wire)	Probe_Wire_Harness	16 req 16 buy	Specs: 10K Ohm NTC Thermistor Temperature Sensor Probe with 1m wire lead Alts/Notes: ETStore SKU ET5524, NTC 10K with steel cap (ETD11498), 100K 3950	[ ]	Est: ₹14.6 [ ]
[ ]	External	J31	SHUNT KELVIN	1x02_P2.54mm_Vertical	1 req 1 buy	Specs: 2.54mm Pitch Breakaway Pin Header (SHUNT KELVIN) Alts/Notes: 40-pin 2.54mm Male/Female header strips	[ ]	Est: ₹5.0 [ ]
	SUBSYSTEM : 01 CELL AFE (13 items)							
[ ]	01 Cell AFE	D1, D2	BAV21WS	D_SOD-323_HandSolder..	2 req 5 buy	Specs: 250V 200mA Fast Switching Diode, SOD-323 Alts/Notes: 1N4148W, BAV19WS, BAV20WS	[ ]	Est: ₹2.5 [ ]
[ ]	01 Cell AFE	R1..R17	20 Ω	R_1206_3216Metric_Pa..	17 req 20 buy	Specs: SMD Chip Resistor 20 Ω, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 20 Ω (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	01 Cell AFE	R18, R21, R22	100 Ω	R_1206_3216Metric_Pa..	3 req 10 buy	Specs: SMD Chip Resistor 100 Ω, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 100 Ω (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	01 Cell AFE	R19..R58	10 kΩ	R_1206_3216Metric_Pa..	22 req 30 buy	Specs: SMD Chip Resistor 10 kΩ, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 10 kΩ (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	01 Cell AFE	R69	300 Ω	R_1206_3216Metric_Pa..	1 req 10 buy	Specs: SMD Chip Resistor 300 Ω, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 300 Ω (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	01 Cell AFE	R70	10 MΩ	R_1206_3216Metric_Pa..	1 req 10 buy	Specs: SMD Chip Resistor 10 MΩ, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 10 MΩ (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	01 Cell AFE	C1..C17	220 nF	C_1206_3216Metric_Pa..	17 req 20 buy	Specs: 220 nF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
[ ]	01 Cell AFE	C18, C20	1 µF	C_1812_4532Metric_Pa..	2 req 10 buy	Specs: 1 µF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
[ ]	01 Cell AFE	C19	100 pF	C_0805_2012Metric_Pa..	1 req 10 buy	Specs: 100 pF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
[ ]	01 Cell AFE	C21	2.2 µF	C_0805_2012Metric_Pa..	1 req 10 buy	Specs: 2.2 µF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
[ ]	01 Cell AFE	C22	100 nF	C_0805_2012Metric_Pa..	1 req 10 buy	Specs: 100 nF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
[ ]	01 Cell AFE	C23	22 nF	C_0805_2012Metric_Pa..	1 req 10 buy	Specs: 22 nF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
[ ]	01 Cell AFE	C24	1 µF	C_0805_2012Metric_Pa..	1 req 10 buy	Specs: 1 µF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
	SUBSYSTEM : 02 POWER SUPPLY (33 items)							
[ ]	02 Power Supply	U11	LM2596S-3.3	TO-263-5_TabPin3	1 req 1 buy	Specs: 3A Step-Down Voltage Regulator (LM2596S-3.3), TO-263-5 Alts/Notes: LM2596S-ADJ + divider, LM2576S, XL4015 module	[ ]	Est: ₹35.0 [ ]
[ ]	02 Power Supply	U12	B1212S-1WR3	Murata_MEE1SxxxxSC_THT	1 req 1 buy	Specs: 1W 12V-to-12V Isolated DC-DC Converter Module Alts/Notes: Mornsun B1212S-1WR3, Hi-Link HLK-B1212S-2WR3	[ ]	Est: ₹103.0 [ ]

[✓]	Subsystem	Ref Des	Component / Value	Package / Footprint	Req / Buy	Technical Specifications & Counter Substitutions	SP Road Store (Write Shop Name)	Quote (₹) (Write Price)
[ ]	02 Power Supply	U13	LM2596S-5.0	TO-263-5_TabPin3	1 req 1 buy	Specs: 3A Step-Down Voltage Regulator (LM2596S-5.0), TO-263-5 Alts/Notes: LM2596S-ADJ + divider, LM2576S, XL4015 module	[ ]	Est: ₹35.0 [ ]
[ ]	02 Power Supply	D4	B340	D_SMC	1 req 2 buy	Specs: 40V 3A Schottky Barrier Rectifier Diode, SMC Alts/Notes: SS34, SK34, MBRS340	[ ]	Est: ₹6.0 [ ]
[ ]	02 Power Supply	D5	B340	D_SMC	1 req 2 buy	Specs: 40V 3A Schottky Barrier Rectifier Diode, SMC Alts/Notes: SS34, SK34, MBRS340	[ ]	Est: ₹6.0 [ ]
[ ]	02 Power Supply	D6	B5819W	D_SMA	1 req 5 buy	Specs: 40V 1A Surface Mount Schottky Barrier Diode, SOD-123 Alts/Notes: 1N5819HW, SS14, BAT54	[ ]	Est: ₹3.0 [ ]
[ ]	02 Power Supply	F1	2A 80V	Fuse_1812_4532Metric	1 req 2 buy	Specs: 2A 80V Fast-Acting Surface Mount Fuse, 1812 Alts/Notes: Bel Fuse 0685P, Littelfuse 0451	[ ]	Est: ₹16.0 [ ]
[ ]	02 Power Supply	F2	3.15A	Fuse_1812_4532Metric	1 req 2 buy	Specs: 3.15A Fast-Acting Surface Mount Fuse, 1812 Alts/Notes: Bel Fuse 0685P, Littelfuse 0451	[ ]	Est: ₹16.0 [ ]
[ ]	02 Power Supply	J1	FAN1	MKDS-1,5-2-5.08_1x02..	1 req 1 buy	Specs: 2-Pin 5.08mm Pitch PCB Screw Terminal Block Alts/Notes: Phoenix MKDS 1,5/2-5.08, Degson DG301-5.08	[ ]	Est: ₹12.0 [ ]
[ ]	02 Power Supply	J2	FAN2	MKDS-1,5-2-5.08_1x02..	1 req 1 buy	Specs: 2-Pin 5.08mm Pitch PCB Screw Terminal Block Alts/Notes: Phoenix MKDS 1,5/2-5.08, Degson DG301-5.08	[ ]	Est: ₹12.0 [ ]
[ ]	02 Power Supply	J4	12V_EXT	MKDS-1,5-2-5.08_1x02..	1 req 1 buy	Specs: 2-Pin 5.08mm Pitch PCB Screw Terminal Block Alts/Notes: Phoenix MKDS 1,5/2-5.08, Degson DG301-5.08	[ ]	Est: ₹12.0 [ ]
[ ]	02 Power Supply	L1	68 uH	L_Bourns_SRR1260	1 req 1 buy	Specs: 68 uH High-Current Shielded Power Inductor (SRR1260) Alts/Notes: Bourns SRR1260, Wurth WE-PDF, Coilcraft MSS1260	[ ]	Est: ₹28.0 [ ]
[ ]	02 Power Supply	L2, L3	33 uH	L_Bourns_SRR1260	2 req 2 buy	Specs: 33 uH High-Current Shielded Power Inductor (SRR1260) Alts/Notes: Bourns SRR1260, Wurth WE-PDF, Coilcraft MSS1260	[ ]	Est: ₹28.0 [ ]
[ ]	02 Power Supply	R101	1M	R_1206_3216Metric	1 req 10 buy	Specs: SMD Chip Resistor 1M, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 1M (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	02 Power Supply	R102	51k	R_0805_2012Metric	1 req 10 buy	Specs: SMD Chip Resistor 51k, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 51k (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	02 Power Supply	R103, R121, R126, R127	100k	R_0805_2012Metric	4 req 10 buy	Specs: SMD Chip Resistor 100k, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 100k (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	02 Power Supply	R104	270k	R_0805_2012Metric	1 req 10 buy	Specs: SMD Chip Resistor 270k, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 270k (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	02 Power Supply	R105	30k	R_0805_2012Metric	1 req 10 buy	Specs: SMD Chip Resistor 30k, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 30k (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	02 Power Supply	R106	470k	R_0805_2012Metric	1 req 10 buy	Specs: SMD Chip Resistor 470k, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 470k (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	02 Power Supply	C27	4.7uF 100V	C_1210_3225Metric	1 req 10 buy	Specs: 4.7uF 100V Multi-layer Ceramic Capacitor (MLCC), 1206/0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹3.5 [ ]
[ ]	02 Power Supply	C28	100nF 100V	C_0805_2012Metric	1 req 5 buy	Specs: 100nF 100V 100V High-Voltage Ceramic Capacitor Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹4.5 [ ]
[ ]	02 Power Supply	C29	2.2nF 50V	C_0805_2012Metric	1 req 10 buy	Specs: 2.2nF 50V 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
[ ]	02 Power Supply	C30	3.3nF	C_0805_2012Metric	1 req 10 buy	Specs: 3.3nF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
[ ]	02 Power Supply	C31	56pF	C_0805_2012Metric	1 req 10 buy	Specs: 56pF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
[ ]	02 Power Supply	C32	22uF 25V	C_1206_3216Metric	1 req 10 buy	Specs: 22uF 25V Multi-layer Ceramic Capacitor (MLCC), 1206/0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹3.5 [ ]
[ ]	02 Power Supply	C33, C35	100nF 25V	C_0805_2012Metric	2 req 10 buy	Specs: 100nF 25V 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
[ ]	02 Power Supply	C34	470uF 25V	CP_Elec_8x10	1 req 2 buy	Specs: 470uF 25V Radial Aluminum Electrolytic Capacitor (Low-ESR) Alts/Notes: Panasonic FR/FC series, Nichicon HE	[ ]	Est: ₹8.5 [ ]
[ ]	02 Power Supply	C36	220uF 10V	CP_Elec_8x10	1 req 2 buy	Specs: 220uF 10V Radial Aluminum Electrolytic Capacitor (Low-ESR) Alts/Notes: Panasonic FR/FC series, Nichicon HE	[ ]	Est: ₹8.5 [ ]
[ ]	02 Power Supply	C37	100nF 10V	C_0805_2012Metric	1 req 10 buy	Specs: 100nF 10V 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
[ ]	02 Power Supply	C38, C39	4.7uF 25V	C_0805_2012Metric	2 req 10 buy	Specs: 4.7uF 25V Multi-layer Ceramic Capacitor (MLCC), 1206/0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹3.5 [ ]
[ ]	02 Power Supply	C40	22uF 25V	C_1206_3216Metric	1 req 10 buy	Specs: 22uF 25V Multi-layer Ceramic Capacitor (MLCC), 1206/0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹3.5 [ ]

[✓]	Subsystem	Ref Des	Component / Value	Package / Footprint	Req / Buy	Technical Specifications & Counter Substitutions	SP Road Store (Write Shop Name)	Quote (₹) (Write Price)
[ ]	02 Power Supply	C41	220uF 10V	CP_Elec_8x10	1 req 2 buy	Specs: 220uF 10V Radial Aluminum Electrolytic Capacitor (Low-ESR) Alts/Notes: Panasonic FR/FC series, Nichicon HE	[ ]	Est: ₹8.5 [ ]
[ ]	02 Power Supply	C42	100nF	C_0805_2012Metric	1 req 10 buy	Specs: 100nF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
	SUBSYSTEM : 03 MCU CONTROL (13 items)							
[ ]	03 MCU Control	U20	CH32V203C8T6	LQFP-48_7x7mm_P0.5mm	1 req 1 buy	Specs: QingKe 32-bit RISC-V MCU, 64KB Flash, 20KB SRAM, LQFP-48 Alts/Notes: Robu.in WeAct CH32V203C8T6 Board (₹199), Evelta STM32F103C8T6 (₹210) Bare MCU: Rare OTC. If not at Vishal, use confirmed ET Store order.	[ ]	Est: ₹95.0 [ ]
[ ]	03 MCU Control	D10	GREEN	LED_0805_2012Metric	1 req 5 buy	Specs: SMD Green Indicator LED, 0805 Alts/Notes: Standard 0805 SMD LEDs (Red/Blue/Green)	[ ]	Est: ₹2.0 [ ]
[ ]	03 MCU Control	J10	WCH-LinkE	1x05_P2.54mm_Vertical	1 req 1 buy	Specs: 2.54mm Pitch Breakaway Pin Header (WCH-LinkE) Alts/Notes: 40-pin 2.54mm Male/Female header strips	[ ]	Est: ₹5.0 [ ]
[ ]	03 MCU Control	J11	Debug UART	1x03_P2.54mm_Vertical	1 req 1 buy	Specs: 2.54mm Pitch Breakaway Pin Header (Debug UART) Alts/Notes: 40-pin 2.54mm Male/Female header strips	[ ]	Est: ₹5.0 [ ]
[ ]	03 MCU Control	J12	OLED Display (I2C)	1x04_P2.54mm_Vertical	1 req 1 buy	Specs: 0.96 inch 128x64 I2C OLED Display Module (SSD1306, 4-Pin) Alts/Notes: SSD1306 0.96 I2C Blue/White, SH1106 1.3 inch I2C OLED	[ ]	Est: ₹171.7 [ ]
[ ]	03 MCU Control	FB10	120R/1.5A	L_0805_2012Metric	1 req 5 buy	Specs: 120 Ohm @ 100MHz 1.5A Ferrite Bead, 0805 Alts/Notes: Murata BLM21PG121SN1D, Sunlord 0805 Bead	[ ]	Est: ₹3.5 [ ]
[ ]	03 MCU Control	Y1	8MHz	Crystal_SMD_3225-4Pi..	1 req 1 buy	Specs: 8.000 MHz Fundamental SMD Crystal Resonator (3225-4P) Alts/Notes: Abracon ABM8, TXC 7M-8.000MAAJ-T, HC-49SMD	[ ]	Est: ₹15.0 [ ]
[ ]	03 MCU Control	R125	1k	R_0805_2012Metric	1 req 10 buy	Specs: SMD Chip Resistor 1k, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 1k (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	03 MCU Control	R128, R129	4.7k	R_0805_2012Metric	2 req 10 buy	Specs: SMD Chip Resistor 4.7k, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 4.7k (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	03 MCU Control	C53	4.7uF	C_0805_2012Metric	1 req 10 buy	Specs: 4.7uF Multi-layer Ceramic Capacitor (MLCC), 1206/0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹3.5 [ ]
[ ]	03 MCU Control	C55	1uF	C_0805_2012Metric	1 req 10 buy	Specs: 1uF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
[ ]	03 MCU Control	C56	10nF	C_0805_2012Metric	1 req 10 buy	Specs: 10nF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
[ ]	03 MCU Control	C57, C58	18pF	C_0805_2012Metric	2 req 10 buy	Specs: 18pF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
	SUBSYSTEM : 04 CURRENT SWITCHING (9 items)							
[ ]	04 Current Switc..	U4, U5, U6, U7, U40	EL817	SOP-4_3.8x4.1mm_P2.5..	5 req 6 buy	Specs: Single Channel Phototransistor Optocoupler, SOP-4 Alts/Notes: PC817C, EL817S, TLP785, LTV-817	[ ]	Est: ₹6.5 [ ]
[ ]	04 Current Switc..	Q1..Q19	IRFB4110	TO-220-3_Vertical	17 req 17 buy	Specs: 100V 180A 3.7mR N-Channel Power MOSFET (TO-220 THT) Alts/Notes: HY4008 (80V 200A), IRFB4310, STP180N10F7 Counter Rule: 100V 180A 3.7mΩ TO-220. Check laser marking; reject cheap clones.	[ ]	Est: ₹56.0 [ ]
[ ]	04 Current Switc..	Q17, Q18	BC817-40,215	SOT-23_Handsoldering	2 req 5 buy	Specs: 45V 500mA General Purpose NPN Transistor, SOT-23 Alts/Notes: MMBT3904, 2N3904, BC847	[ ]	Est: ₹3.0 [ ]
[ ]	04 Current Switc..	R25..R40	27 Ω	R_1206_3216Metric_Pa..	16 req 20 buy	Specs: SMD Chip Resistor 27 Ω, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 27 Ω (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	04 Current Switc..	R59, R61	2.2 kΩ	R_1206_3216Metric_Pa..	2 req 10 buy	Specs: SMD Chip Resistor 2.2 kΩ, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 2.2 kΩ (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	04 Current Switc..	R63, R65	100 kΩ	R_1206_3216Metric_Pa..	2 req 10 buy	Specs: SMD Chip Resistor 100 kΩ, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 100 kΩ (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	04 Current Switc..	R67, R68, R71, R72	330 Ω	R_1206_3216Metric_Pa..	4 req 10 buy	Specs: SMD Chip Resistor 330 Ω, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 330 Ω (Yageo / Walsin / UniOhm)	[ ]	Est: ₹0.8 [ ]
[ ]	04 Current Switc..	C_DRV1, C_DRV4	100 nF	SMD	2 req 10 buy	Specs: 100 nF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹1.2 [ ]
[ ]	04 Current Switc..	C_DRV2, C_DRV3	4.7 µF	SMD	2 req 10 buy	Specs: 4.7 µF Multi-layer Ceramic Capacitor (MLCC), 1206/0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[ ]	Est: ₹3.5 [ ]
	SUBSYSTEM : 05 TEMPERATURE SENSING (3 items)							
[ ]	05 Temperature S..	U8, U9, U30	CD74HC4067SM	SSOP-24_5.3x8.2mm_P0..	3 req 3 buy	Specs: High-Speed 16-Channel Analog Multiplexer / Demultiplexer Alts/Notes: 74HC4067, 74HCT4067, ADG706 (Robu.in SKU 16345, Sunrom)	[ ]	Est: ₹56.9 [ ]

[✓]	Subsystem	Ref Des	Component / Value	Package / Footprint	Req / Buy	Technical Specifications & Counter Substitutions	SP Road Store (Write Shop Name)	Quote (₹) (Write Price)
[ ]	05 Temperature S..	R62..R110	10 kΩ	R_0805_2012Metric_Pa..	32 req 40 buy	Specs: SMD Chip Resistor 10 kΩ, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 10 kΩ (Yageo / Walsin / UniOhm)	[_____]	Est: ₹0.8 [ ]
[ ]	05 Temperature S..	C25, C26	100 nF	C_1206_3216Metric_Pa..	2 req 10 buy	Specs: 100 nF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[_____]	Est: ₹1.2 [ ]
	SUBSYSTEM : 06 BALANCE STATUS (2 items)							
[ ]	06 Balance Status	U31..U34	LTV-847S	SMDIP-16_W9.53mm	4 req 4 buy	Specs: Lite-On 4-Channel DC Optocoupler Isolator (Quad) Alts/Notes: PC847, EL847, TLP281-4, 4x PC817	[_____]	Est: ₹60.5 [ ]
[ ]	06 Balance Status	R130	47k	R_0805_2012Metric	1 req 10 buy	Specs: SMD Chip Resistor 47k, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 47k (Yageo / Walsin / UniOhm)	[_____]	Est: ₹0.8 [ ]
	SUBSYSTEM : 07 ISOLATED CAN & COMMS (11 items)							
[ ]	07 Isolated CAN ..	U15	AMS1117-5.0	SOT-223-3_TabPin2	1 req 1 buy	Specs: 1A Low Dropout Linear Voltage Regulator 5.0V, SOT-223 Alts/Notes: LM1117-5.0, AP1117-50, LD1117S50TR	[_____]	Est: ₹8.0 [ ]
[ ]	07 Isolated CAN ..	F10	1.0A 1812 PTC	Fuse_1812_4532Metric	1 req 2 buy	Specs: 1.0A 16V Resettable Polymeric PTC Fuse, 1812 Alts/Notes: Bourns MF-MSMF110, Littelfuse 1812L110	[_____]	Est: ₹18.0 [ ]
[ ]	07 Isolated CAN ..	J4	PACK LINK, NOT ETHERNET	RJ45_Amphenol_54602-..	1 req 1 buy	Specs: RJ45 8P8C Right-Angle Shielded PCB Modular Jack Alts/Notes: Amphenol 54602-x08, Standard PCB RJ45 Jack	[_____]	Est: ₹25.0 [ ]
[ ]	07 Isolated CAN ..	R107, R120, R131	10k	R_0805_2012Metric	3 req 10 buy	Specs: SMD Chip Resistor 10k, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 10k (Yageo / Walsin / UniOhm)	[_____]	Est: ₹0.8 [ ]
[ ]	07 Isolated CAN ..	R108	2k	R_0805_2012Metric	1 req 10 buy	Specs: SMD Chip Resistor 2k, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 2k (Yageo / Walsin / UniOhm)	[_____]	Est: ₹0.8 [ ]
[ ]	07 Isolated CAN ..	R109	120	R_1206_3216Metric	1 req 10 buy	Specs: SMD Chip Resistor 120, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 120 (Yageo / Walsin / UniOhm)	[_____]	Est: ₹0.8 [ ]
[ ]	07 Isolated CAN ..	C40, C41	100nF	C_0805_2012Metric	2 req 10 buy	Specs: 100nF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[_____]	Est: ₹1.2 [ ]
[ ]	07 Isolated CAN ..	C42	4.7uF 25V	C_0805_2012Metric	1 req 10 buy	Specs: 4.7uF 25V Multi-layer Ceramic Capacitor (MLCC), 1206/0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[_____]	Est: ₹3.5 [ ]
[ ]	07 Isolated CAN ..	C43..C71	100nF	C_0805_2012Metric	12 req 20 buy	Specs: 100nF 50V Ceramic Decoupling Capacitor, 0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[_____]	Est: ₹1.2 [ ]
[ ]	07 Isolated CAN ..	C44	10uF 25V	C_1206_3216Metric	1 req 10 buy	Specs: 10uF 25V Multi-layer Ceramic Capacitor (MLCC), 1206/0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[_____]	Est: ₹3.5 [ ]
[ ]	07 Isolated CAN ..	C46	22uF 10V	C_1206_3216Metric	1 req 10 buy	Specs: 22uF 10V Multi-layer Ceramic Capacitor (MLCC), 1206/0805 Alts/Notes: Standard SMD MLCC (Murata / Samsung / TDK)	[_____]	Est: ₹3.5 [ ]
	SUBSYSTEM : 08 PRECHARGE CIRCUIT (9 items)							
[ ]	08 Precharge Cir..	D41	BZT52C15	D_SOD-123	1 req 5 buy	Specs: 15V 500mW SMD Zener Diode (Gate Clamp, SOD-123) Alts/Notes: MMSZ5245B, BZX384-C15, 15V 500mW Zener SOD-123	[_____]	Est: ₹1.8 [ ]
[ ]	08 Precharge Cir..	J35	AK300-2	MKDS-1,5-2-5.08_1x02..	1 req 1 buy	Specs: 2-Pin 5.08mm Pitch Precharge Resistor Terminal Block (Chassis Resistor Header) Alts/Notes: Phoenix MKDS 1,5/2-5.08, Degson DG301-5.08	[_____]	Est: ₹12.0 [ ]
[ ]	08 Precharge Cir..	R90	220	R_1206_3216Metric	1 req 10 buy	Specs: SMD Chip Resistor 220, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 220 (Yageo / Walsin / UniOhm)	[_____]	Est: ₹0.8 [ ]
[ ]	08 Precharge Cir..	R91	100k	R_1206_3216Metric	1 req 10 buy	Specs: SMD Chip Resistor 100k, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 100k (Yageo / Walsin / UniOhm)	[_____]	Est: ₹0.8 [ ]
[ ]	08 Precharge Cir..	R92	100	R_1206_3216Metric	1 req 10 buy	Specs: SMD Chip Resistor 100, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 100 (Yageo / Walsin / UniOhm)	[_____]	Est: ₹0.8 [ ]
[ ]	08 Precharge Cir..	R93	10k	R_1206_3216Metric	1 req 10 buy	Specs: SMD Chip Resistor 10k, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 10k (Yageo / Walsin / UniOhm)	[_____]	Est: ₹0.8 [ ]
[ ]	08 Precharge Cir..	R94, R96	84.5k 1%	R_1206_3216Metric	2 req 10 buy	Specs: SMD Chip Resistor 84.5k 1%, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 84.5k 1% (Yageo / Walsin / UniOhm) Divider: If 84.5k is missing, use 82k 1% metal film (firmware adjusted).	[_____]	Est: ₹0.8 [ ]
[ ]	08 Precharge Cir..	R95	10k 1%	R_1206_3216Metric	1 req 10 buy	Specs: SMD Chip Resistor 10k 1%, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 10k 1% (Yageo / Walsin / UniOhm)	[_____]	Est: ₹0.8 [ ]
[ ]	08 Precharge Cir..	R97	1k	R_0805_2012Metric	1 req 10 buy	Specs: SMD Chip Resistor 1k, 1% tolerance Alts/Notes: Standard 1% SMD Resistor 1k (Yageo / Walsin / UniOhm)	[_____]	Est: ₹0.8 [ ]

SUBSYSTEM REFERENCE : PCB COPPER FEATURES (NET TIES & SOLDER JUMPERS) - ZERO PROCUREMENT COST

[✓]	Subsystem	Ref Des	Component / Value	Package / Footprint	Req / Buy	Technical Specifications & Counter Substitutions	SP Road Store (Write Shop Name)	Quote (₹) (Write Price)
	SUBSYSTEM : PCB COPPER FEATURES (NET TIES & SOLDER JUMPERS) (8 items)							
[X]	**PCB Copper**	**JP1**	**JP_OPEN**	SolderJumper-2_P1.3m..	**1 req 0 buy**	Specs: 2-Pin Solder Jumper (JP_OPEN) Alts/Notes: Solder bridge / 0-Ohm 0805 resistor PCB Copper Artwork: Manufactured directly on PCB layout. Zero procurement.	ON-BOARD	₹ 0.00
[X]	**PCB Copper**	**JP1**	**TERMINATION**	SolderJumper-2_P1.3m..	**1 req 0 buy**	Specs: 2-Pin Solder Jumper (TERMINATION) Alts/Notes: Solder bridge / 0-Ohm 0805 resistor PCB Copper Artwork: Manufactured directly on PCB layout. Zero procurement.	ON-BOARD	₹ 0.00
[X]	**PCB Copper**	**JP10**	**BOOT0**	SolderJumper-2_P1.3m..	**1 req 0 buy**	Specs: 2-Pin Solder Jumper (BOOT0) Alts/Notes: Solder bridge / 0-Ohm 0805 resistor PCB Copper Artwork: Manufactured directly on PCB layout. Zero procurement.	ON-BOARD	₹ 0.00
[X]	**PCB Copper**	**JP11**	**ADDR0**	SolderJumper-2_P1.3m..	**1 req 0 buy**	Specs: 2-Pin Solder Jumper (ADDR0) Alts/Notes: Solder bridge / 0-Ohm 0805 resistor PCB Copper Artwork: Manufactured directly on PCB layout. Zero procurement.	ON-BOARD	₹ 0.00
[X]	**PCB Copper**	**JP12**	**ADDR1**	SolderJumper-2_P1.3m..	**1 req 0 buy**	Specs: 2-Pin Solder Jumper (ADDR1) Alts/Notes: Solder bridge / 0-Ohm 0805 resistor PCB Copper Artwork: Manufactured directly on PCB layout. Zero procurement.	ON-BOARD	₹ 0.00
[X]	**PCB Copper**	**JP13**	**ADDR2**	SolderJumper-2_P1.3m..	**1 req 0 buy**	Specs: 2-Pin Solder Jumper (ADDR2) Alts/Notes: Solder bridge / 0-Ohm 0805 resistor PCB Copper Artwork: Manufactured directly on PCB layout. Zero procurement.	ON-BOARD	₹ 0.00
[X]	**PCB Copper**	**JP14**	**EE_WP**	SolderJumper-2_P1.3m..	**1 req 0 buy**	Specs: 2-Pin Solder Jumper (EE_WP) Alts/Notes: Solder bridge / 0-Ohm 0805 resistor PCB Copper Artwork: Manufactured directly on PCB layout. Zero procurement.	ON-BOARD	₹ 0.00
[X]	**PCB Copper**	**NT1, NT2, NT3**	**NetTie_2**	NetTie-2_SMD_Pad0.5mm	**3 req 0 buy**	Specs: Net Tie Trace Link (Copper feature on PCB) Alts/Notes: Direct PCB trace copper polygon PCB Copper Artwork: Manufactured directly on PCB layout. Zero procurement.	ON-BOARD	₹ 0.00

16S BMS PROCUREMENT RECAP & VENDOR COST AUDIT:
• Part 1 (Non-ET Store Items): 21 Line Items | Design Qty: 22 pcs | Order Qty: 34 pcs | Baseline: ₹ 2,887.91
Vendors: Rajiv Electronics (7 items, ₹448.10), Evelta (7 items, ₹814.64), Robu.in (5 items, ₹775.00), LionCircuits (1 item, ₹665.29), Sharvi (1 item, ₹184.88).
• Part 2 (ET Store Sourced Items): 103 Line Items | Design Qty: 280 pcs | Order Qty: 785 pcs | Baseline: ₹ 3,407.92
Includes: 17x IRFB4110 MOSFETs, CH32V203 bare MCU, 3x CD74HC4067, 5x EL817, 4x LTV-847S, 0.96" OLED, inductors, 1% SMD resistors, 50V/100V MLCC caps.
• Grand Turnkey Total: 124 Line Items | 302 Design Pcs | 819 Commercial Order Pcs | Total: ₹ 6,295.83

ON-SITE ASSEMBLY & HARDWARE VERIFICATION CHECKLIST:
[ ] Precharge Resistor Verified: 50 Ω 25W aluminum chassis resistor (pitch 55mm) in hand. Connects to J35.
[ ] High-Voltage MLCC Caps: Verify 100V rating for C18, C20 (AFE top of pack) and C27, C28 (buck input).
[ ] Gate Drive Bias: Confirm B1209S provides isolated +9V rail referenced to FET_SOURCE_COMMON.
[ ] Resistor Tolerances: Check 1% markings on divider chains (R94, R96 84.5k and R62..R110 10k).
[ ] High-Current Hardware: Direct bolt-down M5/M6 brass/SS hardware for J32, J33, J34 onto copper busbar.